

MOBILE PHONE SALES ANALYSIS

BY

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THE DATA IMMERSED EXCEL CAPSTONE PROJECT

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SUMMARY

This project presents a comprehensive sales performance analysis for Velocity Phone Inc., using Microsoft Excel to transform raw transactional mobile phone sales data into actionable business intelligence dashboards. In this project, I analyzed 366 transaction records, which were reduced to 303 clean and valid records after duplicate removal and data preparation.

I developed two interactive dashboards:

- Dashboard 1: Sales Performance Overview
- Dashboard 2: Customer and Product Analysis

The analysis provided critical insights into:

- Country-level sales performance
- Brand revenue contribution
- Customer demographic purchasing behavior
- Product performance
- Pricing strategy effectiveness
- Sales channel performance

The dashboards provide management with data-driven insights to support strategic decision-making and revenue optimization.

PROJECT BACKGROUND

As a Data Analyst for Velocity Phone Inc., I was tasked to design a Mobile Phone Sales Analysis Dashboard(s) in Microsoft Excel that will identify sales trends, assess product performance across different markets, and support decision-making for strategic business initiatives.

Velocity Phone Inc. operates in the competitive smartphone market across multiple countries, offering several mobile brands, storage variants, and sales channels. To improve decision-making, management requires a centralized analytical reporting system that can monitor:

- Sales performance
- Customer Behavior
- Product Performance

To conduct this analysis, I utilized Microsoft Excel, using features such as pivot tables, pivot charts, and slicers to develop an interactive Excel dashboards to uncover patterns and derive actionable insights.

PROJECT OBJECTIVES

The specific objectives of this project are:

1. Problem Statement Definition

To identify and clearly define the key business challenges affecting Velocity Phone Inc. based on insights revealed from the dataset, including variations in sales performance across countries, customer segments, product categories, and sales channels.

2. Data Cleaning and Transformation

To examine, clean, and prepare the raw dataset for analysis by removing duplicate records, ensuring data consistency, handling missing values where necessary, and creating calculated columns to enrich the dataset for deeper analytical insights.

3. Data Analysis

To analyze the dataset in order to uncover meaningful sales trends and patterns across product attributes, customer demographics, geographic locations, sales channels, and pricing categories.

This includes identifying:

- Sales variations across countries
- Customer purchasing behavior across age groups and gender
- Product performance across mobile models and storage capacities
- Relationships between pricing categories, units sold, and total revenue
- Sales channel performance and contribution to total sales

4. Data Visualization and Dashboard Development

To design and develop interactive Mobile Phone Sales Analysis Dashboards in Microsoft Excel using pivot tables, pivot charts, KPI cards, and slicers that allow dynamic exploration of sales data and support effective performance monitoring and decision-making.

5. Sales Performance and Customer Insights

To identify high-performing and underperforming business segments, evaluate customer preferences, and generate actionable insights that explain performance trends across products, regions, and customer demographics.

6. Business Strategy and Recommendations

To provide data-driven recommendations that support Velocity Phone Inc. in improving inventory planning, strengthening sales channel strategies, optimizing market focus, enhancing customer targeting, and improving overall business performance.

PROBLEM STATEMENT

Velocity Phone Inc. is a major mobile phone retailer operating across multiple countries in Asia. The company is seeking deeper insights into sales performance, customer behavior, and product trends to optimize inventory management, improve marketing strategies, and enhance overall business performance.

Despite having access to transactional sales data, The Company lacked a structured analytical system for effectively monitoring mobile phone sales performance across key business dimensions. Without structured analysis, it becomes difficult to:

- Identify top-performing markets
- Track monthly revenue trends
- Understand customer preferences
- Evaluate pricing performance
- Optimize product inventory allocation
- Improve sales strategy

The key challenge facing Velocity Phone Inc. is the need to transform raw transactional data into meaningful business intelligence that supports effective sales monitoring, customer analysis, product performance evaluation, and data-driven strategic planning & decision making.

To address this challenge, I developed the Mobile Phone Sales Analysis Dashboards to provide an interactive and comprehensive analytical solution for tracking performance trends and supporting informed business decisions.

DATA CLEANING AND TRANSFORMATION

Dataset Description

The dataset used for this project contains transactional sales records for mobile phone purchases across multiple countries in Asia for Velocity Phone Inc. in 2024.

It captures detailed information relating to product specifications, customer demographics, transaction details, sales channels, geographic locations, and payment methods. The dataset was designed to support sales performance analysis across multiple business dimensions.

The original dataset contained 366 transaction records (367 including the header) and multiple variables describing each transaction. After the data cleaning process, the dataset was reduced to 303 valid records following the removal of duplicate entries, resulting in a cleaner and reliable dataset for analysis.

Dataset Structure

The dataset was provided in an Excel workbook containing two sheets:

- **Fact Sales** – This is the primary dataset sheet containing the transactional sales records used for analysis.
- **Data Dictionary** – This sheet provides descriptions for each column in the dataset (like validating the uniqueness of the Transaction ID field).

The key fields in the dataset include:

- Transaction details (transaction id, transaction date, end of month, transaction month)
- Product information (brand, mobile model, storage size, color, operating system, price, price evaluation)
- Sales metrics (units sold, sales)
- Customer demographics (customer age, customer age-group, customer gender)
- Sales operations (sales channel, payment type)
- Geographic information (country, city, latitude, longitude)

Data Cleaning

Several cleaning steps were performed to ensure data quality and analytical reliability.

- **Duplicate Removal** – Duplicate records were identified and removed from the dataset. This step was necessary because the Transaction ID column was described as unique. However, duplicate transaction IDs were observed with identical values across all fields except for minor differences in transaction date and end of month entries.
- **Missing Value** – The dataset was checked for blanks and null values across all columns and no missing values were found.

- **Data Formatting** – Currency formatting was applied to the price column and date fields were standardized for accurate time-based analysis. Data categories were reviewed for consistency across text fields.

Data Transformation

To enrich the dataset for analysis, additional calculated columns were created:

- **Transaction Month** – A transaction month column was created from transaction dates to enable monthly trend analysis and identify seasonal sales pattern
- **Sales** – A calculated sales metric was created using: $\text{Sales} = \text{Price} \times \text{Units Sold}$. This supports total sales analysis and measures product and market performance.
- **Price Evaluation** – A Price Evaluation column was created to categorize products into pricing segments:

Low Cost: $\leq \$600$

Mid Cost: $\$601 - \1000

High Cost: $> \$1000$

This enabled pricing strategy analysis like analyzing customer purchasing behavior by price segment, compare unit demand across pricing categories and evaluate sales contribution by product pricing level.

DATA ANALYSIS

Overview of Data Analysis Approach

After completing data cleaning and transformation, the dataset was analyzed using Microsoft Excel Pivot Tables to uncover meaningful patterns, trends, and relationships across key business dimensions. The analysis focused on evaluating sales performance from multiple perspectives, including:

- Geographic performance
- Product performance
- Customer demographics
- Pricing categories
- Sales channel contribution
- Monthly sales trends

I created several PivotTables to support the analytical process and provide the foundation for dashboard development. These pivot tables enabled aggregation, comparison, and summarization of key performance metrics such as total sales revenue and units sold.

Geographic Sales Analysis

This analysis was performed to identify market performance differences across countries. The analysis revealed noticeable differences in revenue contribution across countries, indicating differences in market performance and customer demand across Velocity Phone Inc. operational regions.

Row Labels	Sum of Sales
India	\$5.97M
Turkey	\$4.31M
Bangladesh	\$1.59M
Pakistan	\$0.29M
Grand Total	\$12.16M

Figure 1 Sales by Country

Customer Demographic Analysis

Customer demographic analysis was performed to understand purchasing behavior across different customer segments.

The analysis focused on:

- Age Group Analysis

This examined how purchasing behavior varies across customer age categories to identify the most commercially valuable customer segments.

Row Labels	Sum of Sales
26-33	\$2.39M
42-49	\$2.31M
18-25	\$2.15M
34-41	\$1.88M
58-65	\$1.86M
50-57	\$1.56M
Grand Total	\$12.16M

Figure 2 Sales by Age-Group

➤ Gender Analysis

This evaluated revenue contribution across gender categories to understand differences in purchasing behavior.

Row Labels	Sum of Sales
Female	\$6.09M
Male	\$5.04M
Other	\$1.03M
Grand Total	\$12.16M

Figure 3 Sales by Gender

Product Performance Analysis

Product performance analysis was conducted to evaluate how different mobile products performed across sales and unit volume metrics. The analysis focused on:

➤ Brand Performance

To identify which brands generated the highest overall sales.

Row Labels	Sum of Sales
Samsung	\$3.15M
Apple	\$2.94M
OnePlus	\$2.44M
Google	\$2.13M
Xiaomi	\$1.50M
Grand Total	\$12.16M

Figure 4 Sales by Brand

➤ Mobile Model Performance

To determine the Top-performing mobile models based on sales revenue and units sold.

Row Labels	Sum of Sales
Z Fold 6	\$1.60M
Galaxy S25 Ultra	\$1.14M
Pixel 9 Pro	\$1.10M
OnePlus 12 Pro	\$1.01M
OnePlus 11R	\$0.79M
iPhone 15	\$0.72M
Grand Total	\$6.37M

Figure 5 Top 6 Mobile Model by Sales

Row Labels	Sum of Units_Sold
OnePlus 12 Pro	1,196
OnePlus 11R	1,194
OnePlus Nord 4	1,161
Pixel 9 Pro	1,050
Galaxy S25 Ultra	922
Pixel 9	882
Grand Total	6,405

Figure 6 Top 6 Mobile Model by Units Sold

➤ Storage Capacity Analysis

To evaluate customer preferences for device storage options. This analysis helps identify the storage size that recorded the highest customer demand, the one that generated the highest sales and the relationship between units sold volume and revenue generation across storage categories.

Row Labels	Sum of Sales
128GB	\$4.27M
256GB	\$4.17M
64GB	\$3.72M
Grand Total	\$12.16M

Figure 7 Sales by Storage Size

Row Labels	Sum of Units_Sold
256GB	5,398
64GB	5,090
128GB	4,898
Grand Total	15,386

Figure 8 Units Sold by Storage Size

Pricing Category Analysis

The custom Price Evaluation field enabled pricing segmentation analysis. Products were grouped into:

- High Cost
- Mid Cost
- Low Cost

This analysis was conducted to examine the relationship between Product price level, Units sold and Revenue contribution. The analysis provided insights into customer purchasing preferences across pricing categories.

Row Labels	Sum of Sales
High-Cost	\$5.09M
Mid-Cost	\$4.40M
Low-Cost	\$2.67M
Grand Total	\$12.16M

Figure 9 Sales by Price Evaluation

Row Labels	Sum of Units Sold
Low-Cost	5,839
Mid-Cost	5,569
High-Cost	3,978
Grand Total	15,386

Figure 10 Units Sold by Price Evaluation

Sales Channel Analysis

Sales channel analysis examined performance across the company's available sales channels (Online, Retail Store, and Partner). This analysis was used to evaluate channel contribution to total revenue and identify the most effective sales channel.

Row Labels	Sum of Sales
Online	\$7.49M
Retail Store	\$3.22M
Partner	\$1.44M
Grand Total	\$12.16M

Figure 11 Sales by Sales Channel

Monthly Sales Trend Analysis

Monthly sales analysis was conducted using the generated Transaction Month column.

This analysis helped identify:

- Sales fluctuations over time
- High-performing months
- Low-performing months

Understanding these trends supports improved sales planning and campaign timing.

Row Labels	Sum of Sales
Jan	\$1.41M
Feb	\$1.35M
Mar	\$1.06M
Apr	\$1.04M
May	\$0.96M
Jun	\$0.97M
Jul	\$0.99M
Aug	\$1.10M
Sep	\$0.68M
Oct	\$0.93M
Nov	\$0.60M
Dec	\$1.08M
Grand Total	\$12.16M

Figure 12 Sales by Month

Key Analytical Insights

The data analysis process revealed several important patterns:

- Sales performance varies significantly across countries, with some markets contributing substantially more revenue than others.
- Customer purchasing behavior differs across age groups and gender categories.
- Certain mobile brands and models outperform others in total sales and unit volume.
- Higher-priced devices generate stronger revenue performance despite lower unit sales volume compared to lower-priced categories.
- Online sales channels contribute the highest share of total revenue compared to retail store and partner channels.
- Monthly sales performance fluctuates across the year, indicating variations in customer demand over time.
- Storage size analysis revealed differing performance patterns, with 256GB devices recording the highest unit sales while 128GB devices generated the highest sales revenue.

These analytical insights formed the basis for the dashboard visualizations and subsequent business recommendations.

DATA VISUALIZATION AND DASHBOARD DEVELOPMENT

Overview of Dashboard Development

Following data analysis, the insights obtained from the pivot tables were transformed into two interactive Mobile Phone Sales Analysis Dashboards using Microsoft Excel.

The dashboards were designed to provide a clear, visual representation of key business metrics and performance trends, enabling dynamic exploration of sales data for effective decision-making.

The dashboards development process focused on:

- Presenting complex data in a simplified visual format
- Highlighting key performance indicators (KPIs)
- Supporting comparative analysis across multiple business dimensions
- Enabling interactive data exploration through filtering tools

The dashboards were developed using Microsoft Excel visualization tools, including:

- Pivot Charts
- Pivot Tables
- Key Performance Indicators (KPIs)
- Slicers
- Data Formatting and Layout Design

Dashboard 1: Sales Performance Dashboard

The first dashboard focuses on overall sales performance across key operational dimensions. It was designed to provide insights into revenue trends, market performance, brand contribution, and sales channel effectiveness.

Dashboard 1 includes the following KPI cards:

- Total Sales
- Top Sales Month
- Top Performing Country
- Top Performing Brand

The dashboard contains the following charts:

- Sales by Country
- Sales by Month
- Sales by Brand
- Sales by Sales Channel
- Sales by Price Evaluation

These visualizations provide a high-level overview of overall business sales performance.



Figure 13 Sales Performance Dashboard

Dashboard 2: Customer and Product Analysis Dashboard

The second dashboard focuses on customer behavior and product performance analysis. It was designed to provide insights into purchasing patterns, product demand, and customer segmentation.

Dashboard 2 includes the following KPI Cards:

- Total Units Sold
- Top Mobile Model
- Top Storage Size
- Top Customer Age Group

The dashboard contains the following charts:

- Units Sold by Price Evaluation
- Units Sold by Storage Size
- Sales by Gender
- Sales by Age Group
- Top 6 Mobile Models by Sales
- Top 6 Mobile Models by Units Sold

These visualizations support deeper analysis of customer preferences and product performance.



Figure 14 Customer & Product Analysis Dashboard

Dashboard Interactivity and User Experience

To enhance usability, both dashboards were designed with interactive filtering capabilities through slicers. These slicers allow users to dynamically explore data and customize analysis views based on selected criteria.

The dashboard development process successfully transformed raw transactional data into interactive analytical tools capable of supporting business intelligence and strategic decision-making. The dashboards provide clear visualization of sales trends, real-time analytical exploration, performance monitoring across multiple dimensions, and actionable business insights.

SALES PERFORMANCE AND CUSTOMER INSIGHTS

This section presents insights derived from the two dashboards analysis, focusing on overall sales performance, customer purchasing behavior, product preferences, and market trends. The analysis provides a clearer understanding of the factors influencing revenue generation and customer engagement across different market segments.

Key Performance Indicators (KPIs)

The dashboards provide an overview of the company's overall sales performance and customer and product insights through key performance indicators.

TOTAL SALES	TOP SALES MONTH	TOP COUNTRY	TOP BRAND
\$12.16M	\$1.41M	INDIA	SAMSUNG
TOTAL UNITS SOLD	TOP MOBILE MODEL	TOP STORAGE SIZE	TOP AGE GROUP
15,386	Z Fold 6	256GB	26-33

Figure 15 Key Performance Indicators

The analysis shows that in 2024 Velocity Phone Inc. generated a total sales of \$12.16 million, \$1.41 million as the highest sales in a month, with 15,386 units sold across all product categories. Samsung emerged as the top-performing brand, while the Z Fold 6 recorded the highest sales among individual mobile models. The 256GB storage size was the most preferred storage option, and customers aged 26–33 years represented the most active purchasing segment. India also emerged as the strongest-performing market.

Country Sales Performance

The country sales chart was analyzed to evaluate market performance across different regions.

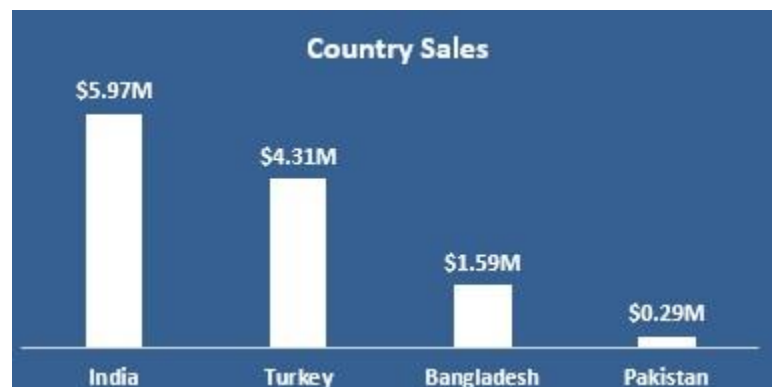


Figure 16 Country Sales Chart

The analysis reveals India as the dominant market with \$5.97M in sales, followed by Turkey at \$4.31M. Bangladesh contributed a moderate \$1.59M and Pakistan contributed significantly less at \$0.29M.

➤ **Insight:**

The strong performance in India and Turkey indicate higher customer acceptance and stronger market penetration in these regions. Lower sales performance in Pakistan indicates lower customer engagement.

Monthly Sales Trend

The monthly sales trend chart was analyzed to evaluate sales consistency throughout the year.



Figure 17 Monthly Sales Chart

The analysis reveals strong sales performance at the beginning of the year, with Jan at \$1.41M and Feb at \$1.35M. Sales then gradually declined through mid-year, hitting a low in Sep at \$0.68M and Nov at \$0.60M. A noticeable recovery was observed toward the end of the year with Dec at \$1.08M.

➤ **Insight:**

Sales varied across months, with stronger performance recorded in the earlier months, indicating uneven revenue distribution throughout the year.

Brand Sales Performance

The brand sales chart provides insight into brand performance.



Figure 18 Brand Sales Chart

The chart shows Samsung leading with \$3.15M, closely followed by Apple at \$2.94M. OnePlus, Google, and Xiaomi followed with \$2.44M, \$2.13M, and \$1.50M respectively.

➤ Insight:

Samsung and Apple emerged as the high-performing brands. The \$1.65M gap between the top and bottom brands shows tiered market positioning, with premium brands capturing higher revenue share.

Sales Channel Performance

The sales channel analysis examined how customers prefer to purchase mobile devices.



Figure 19 Channel Sales Chart

The chart shows that online sales generated the largest share of revenue (\$7.49M), significantly outperforming retail store (\$3.22M) and partner (\$1.44M) channel sales.

➤ **Insight:**

Online sales dominate customer purchasing behavior, this highlights a strong customer preference for digital purchasing channels. Retail stores remain significant, showing some customers still value physical purchase.

Sales by Price Evaluation

The price evaluation chart examined revenue performance across pricing segments.



Figure 20 Sales by Price Evaluation

The chart shows high-cost devices generated the most revenue at \$5.09M, followed by mid-cost at \$4.40M and low-cost at \$2.67M.

➤ **Insight:**

This shows that customers are willing to spend more on premium smartphones, reflecting strong demand for advanced features and high-end device specifications.

Unit Sold by Price Evaluation

The unit sales analysis offers a different perspective by focusing on purchase volume rather than revenue.



Figure 21 Unit Sold by Price Evaluation

The chart reveals low-cost devices led in units sold with 5,839 units, followed by mid-cost at 5,569 and high-cost had the lowest volume at 3,978 units.

➤ **Insight:**

Although premium smartphones generate more revenue, affordability remains a major factor influencing purchase decisions for a larger portion of customers.

Storage Size Performance

The storage analysis examined customer preference across device storage capacities.



Figure 22 Unit sold by Storage Size

The chart shows 256GB as the highest unit sales with 5,398 units sold, followed by 64GB at 5,090 and 128GB at 4,898 units.

➤ **Insight:**

This indicates strong customer preference for larger storage capacities, reflecting increased demand for devices that support heavier application usage and media storage. Further

analysis from the pivot table revealed that 128GB devices generated the highest overall sales despite recording lower unit sales compared to 256GB devices.

Gender-Based Sales Analysis

The gender sales analysis explored sales contribution across customer gender segments.

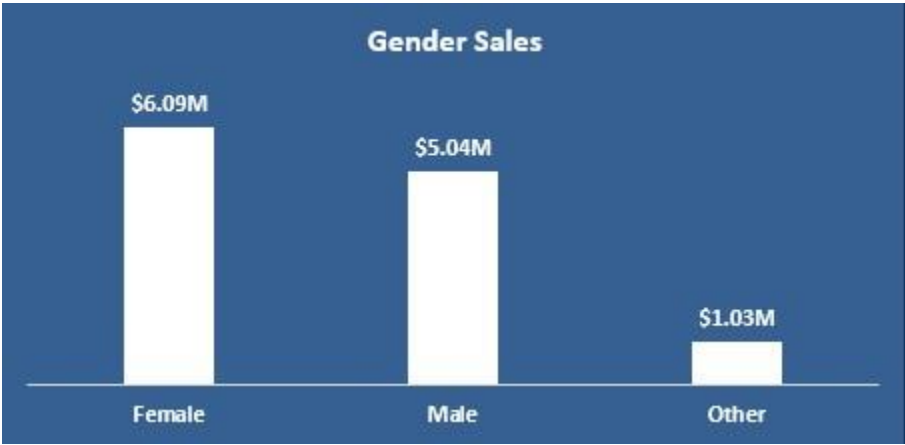


Figure 23 Gender Sales Chart

The chart reveals female customers generated the highest sales at \$6.09M, compared to male at \$5.04M and other at \$1.03M.

➤ **Insight:**

Female customers generated the highest total sales, exceeding male and other customer segments

Age Group Performance

The age-group analysis identified the most valuable customer demographics.

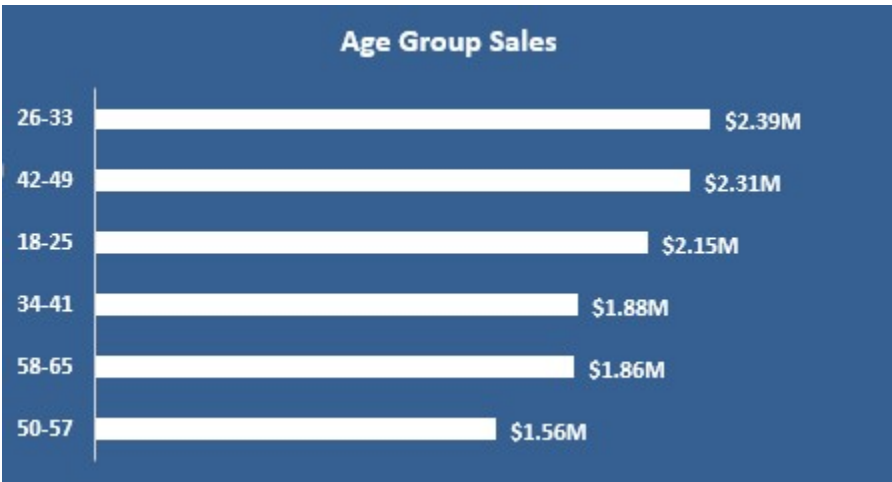


Figure 24 Age-Group Sales Chart

The analysis shows the 26-33 age group leading with \$2.39M in sales, followed by 42-49 at \$2.31M. The 50-57 group contributed the least at \$1.56M.

➤ **Insight:**

Customer aged 26-33 represent the company's most valuable customer segment.

Mobile Model Performance by Sales

The mobile model sales chart identified the top 6 highest revenue-generating products.



Figure 25 Top6 Mobile Model Sales

Samsung models Z Fold 6 (\$1.60M) and Galaxy s25 Ultra (\$1.14M) outperformed all other models in sales, followed by devices from Google, OnePlus, and iPhone.

➤ **Insight:**

Samsung's premium models dominates revenue generation, indicating strong market demand for high-end devices.

Mobile Model Performance by Units Sold

The unit sales chart analyzed the top 6 mobile models by units sold.

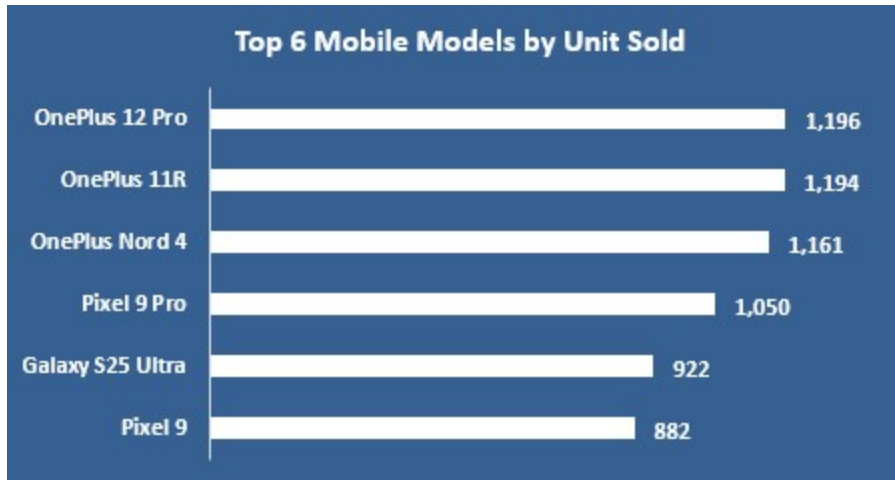


Figure 26 Top 6 Mobile Models by Unit Sold

The analysis shows OnePlus 12 Pro narrowly leading with 1,196 units, followed by OnePlus 11R at 1,194 and OnePlus Nord 4 at 1,161. Volume leaders differ from revenue leaders.

➤ **Insight:**

This suggests that customers strongly value products that balance performance and affordability. While premium devices generate higher revenue, affordably priced models achieve higher purchase volume.

BUSINESS STRATEGY AND RECOMMENDATIONS

Based on the insights from the analysis, the following strategic recommendations are proposed to help Velocity Phone Inc. improve inventory planning, strengthen marketing performance, and expand market reach.

Strengthen Focus on High-Performing Markets

The country sales analysis revealed that India and Turkey generated the highest revenue, significantly outperforming Bangladesh and Pakistan.

➤ Recommendation:

Velocity Phone Inc. should allocate more marketing resources and inventory to India and Turkey to maximize returns in these high-performing markets. At the same time, targeted promotional campaigns should be introduced in lower-performing regions such as Pakistan to stimulate demand and improve market penetration.

Optimize Inventory Based on Product Demand

The dashboard showed that premium models such as the Z Fold 6 generated the highest revenue, while OnePlus models recorded the highest unit sales.

➤ Recommendation:

The company should maintain higher inventory levels for top-selling affordably priced models to meet volume demand, while ensuring sufficient stock of premium devices that contribute strongly to revenue generation.

Increase Investment in Online Sales Channels

The sales channel analysis showed that online sales contributed the largest share of total revenue.

➤ Recommendation:

Velocity Phone Inc. should prioritize digital sales strategies by improving e-commerce platforms, investing in online advertising, offering exclusive online deals and introducing online promotions. This will help capitalize on customer preference for digital purchasing channels.

Develop Pricing Strategies for Different Customer Segments

The analysis showed that high-cost devices generated the highest revenue, while low-cost devices recorded the highest unit sales.

➤ Recommendation:

The company should maintain a balanced pricing strategy by continuing to promote premium devices for revenue growth while also expanding affordable smartphone options to customers seeking affordability.

Target Marketing Toward Key Customer Demographics

The dashboard identified customers aged 26–33 as the most active purchasing segment, while female customers generated the highest revenue contribution.

➤ **Recommendation:**

Marketing campaigns should be designed to target young adult consumers through digital platforms and personalized promotions. Since female customers generated the highest revenue contribution, targeted campaigns designed around this demographic may further improve customer retention and sales performance.

Expand Mid-Range Product Positioning

Storage analysis showed strong demand for 256GB devices, while combined sales insights indicate strong performance for competitively priced models.

➤ **Recommendation:**

Velocity Phone Inc. should expand its mid-range product offerings with larger storage options, as these products provide an effective balance between affordability and premium functionality.

Improve Monthly Sales Stability

The monthly sales trend showed stronger performance at the beginning and end of the year, with noticeable declines during some mid-year periods.

➤ **Recommendation:**

The company should strengthen promotional activities and introduce targeted discounts during low-performing months and double down on marketing campaigns, product bundling and upselling strategies during high-performing months to maximize returns.

Overall, Velocity Phone Inc. should maintain a balanced product strategy that supports both premium high-revenue devices and lower-cost high-volume products. This dual approach will ensure broader market coverage and greater revenue stability.

CONCLUSION

In conclusion, the dashboard analysis demonstrates that Velocity Phone Inc. has significant growth opportunities. By implementing strategic inventory planning, expanding digital channels, adopting customer-focused marketing, and driving product innovation, the company can enhance operational efficiency, increase customer satisfaction, and achieve sustainable revenue growth.